

LAB 1 : Fourier Series and Spectral Analysis

Lab Group:

Lab completion date:

Teacher:

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The questions will be answered directly. Additional curves and explanations can be provided on free paper or better in an `m1x` MATLAB file.

1.1 Discrete Fourier Transform (from Signal processing Toolbox)

The discrete Fourier transform, or DFT, is the primary tool of digital signal processing. The foundation of the product is the fast Fourier transform (FFT), a method for computing the DFT with reduced execution time. Many of the toolbox functions (including Z-domain frequency response, spectrum and cepstrum analysis, and some filter design and implementation functions) incorporate the FFT. The MATLAB environment provides the functions `fft` and `ifft` to compute the discrete Fourier transform and its inverse, respectively. For the input sequence x and its transformed version X (the discrete-time Fourier transform at equally spaced frequencies around the unit circle), the two functions implement the relationships

$$X(k+1) = \sum_{n=0}^{N-1} x(n+1)W_N^{kn}$$

$$x(n+1) = \frac{1}{N} \sum_{k=0}^{N-1} X(k+1)W_N^{-kn}.$$

In these equations, the series subscripts begin with 1 instead of 0 because of the MATLAB vector indexing scheme, and

$$W_N = e^{-j2\pi/N}.$$

Execute and comment the following MATLAB commands

```

1 t = 0:1/100:10-1/100;           % Time vector
2 x = sin(2*pi*15*t) + sin(2*pi*40*t); % Signal
3 y = fft(x);                     % Compute DFT of x
4 m = abs(y);                     % Magnitude
5 y(m<1e-6) = 0;
6 p = unwrap(angle(y));           % Phase

```

To plot the magnitude and phase in degrees, type the following commands:

```

1 f = (0:length(y)-1)*100/length(y); % Frequency vector
2
3 subplot(2,1,1)
4 plot(f,m)
5 title('Magnitude')
6 ax = gca;
7 ax.XTick = [15 40 60 85];
8
9 subplot(2,1,2)
10 plot(f,p*180/pi)
11 title('Phase')
12 ax = gca;
13 ax.XTick = [15 40 60 85];

```

In the following the number of points is specified.

```

1 n = 512;
2 y = fft(x,n);
3 m = abs(y);
4 p = unwrap(angle(y));
5 f = (0:length(y)-1)*100/length(y);
6
7 subplot(2,1,1)
8 plot(f,m)
9 title('Magnitude')
10 ax = gca;
11 ax.XTick = [15 40 60 85];
12 subplot(2,1,2)
13 plot(f,p*180/pi)
14 title('Phase')
15 ax = gca;
16 ax.XTick = [15 40 60 85];

```

The inverse discrete Fourier transform function `ifft` is used to compute back the initial samples.

```

1 t = 0:1/255:1;
2 x = sin(2*pi*120*t);
3 y = real(ifft(fft(x)));
4 figure
5 plot(t,x-y)

```

1.2 Fourier series decomposition

We consider the periodic sawtooth signal of period $T = 1$ sec defined for one period by $s(t) = at \in \left[-\frac{T}{2}, \frac{T}{2}\right]$. The constant is $a = 2$.

1. Draw the signal for $a = 2$ and $t \in [0, 5]$ sec. Generate The signal for the same sampling time values with a sampling period $T_e = 0.02$ sec. Draw the (functions `linspace`, `plot`, ... `xlabel`, `ylabel`).

2. Show that the coefficients a_n of the Fourier series are zero. For which reasons the coefficients b_n are an infinite continuation? What is the mean value of the signal? Verify this value using the function `mean`.

3. Show that the coefficients of the Fourier series are given by

$$b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} at \sin\left(\frac{2\pi}{T}nt\right) dt = \frac{aT}{\pi n} (-1)^{n+1}$$

How does the module of these when coefficients evolve when n approaches $+\infty$.

4. Generate on Matlan the 4 signals, using the same temporal basis than s , $s_1 = b_1 \sin\left(\frac{2\pi}{T}t\right)$, $s_2 = b_2 \sin\left(\frac{2\pi}{T} \times 2 \times t\right)$, $s_3 = b_3 \sin\left(\frac{2\pi}{T} \times 3 \times t\right)$, $s_4 = b_4 \sin\left(\frac{2\pi}{T} \times 4 \times t\right)$. Overlay both signals (s , s_1 , $s_1 + s_2$, $s_1 + s_2 + s_3$, $s_1 + s_2 + s_3 + s_4$). What can you conclude?

5. Using the function `cov`, calculate the power of each signal s , s_1 , s_2 , s_3 , s_4 . Compute the power of the signals ($s_1 + s_2$, $s_1 + s_2 + s_3$, $s_1 + s_2 + s_3 + s_4$). Compare these powers to the coefficients b_n and to the power of signal s . Conclude.

6. Comment the figure obtained by the following Matlab commands

```

1 plot(linspace(-1/(2*Te),1/(2*Te),length(s)),fftshift(abs(fft(s))))
2
3 hold on
4 F=linspace(-1/(2*Te),1/(2*Te),length(s));
5 Spectre=fftshift(abs(fft(s1+s2+s3+s4)));
6 plot(F,Spectre,'r')

```

1.3 Signal spectrum

Consider the signal $x(t) = \exp(-at)u(t)$, a is a positive real and $u(t)$ the unit step function.

1. Draw the signal for $a = 2$ and $t \in [0, 10]$ sec. Generate the signal for the same previous values and a sample time of $T_e = 0.1$ sec. Draw the signal. (functions *linspace*, *exp*, *stem*, *xlabel*, *ylabel*)
2. Calculate the Fourier transform of the signal, noted $X(f)$. Draw the modulus of $X(f)$ and its phase for frequencies in the interval $[-5, 5]$. (functions *sqr*, *plot*). Give the value of amplitude spectrum at a frequency of 1Hz.
3. Draw the amplitude spectrum for the frequencies in the interval $[-5, 5]$. (functions *abs*, *fftshift*, *fft*). Do not forget the normalisation by Fe . Compare numerical values to theoretical ones at a frequency of 1Hz. Explain the difference.

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4. Take back the previous plots for $T_e = 0.5$ sec and $T_e = 0.01$ sec. Conclude.